LOVEINSTORE

Trainer Management Web Panel

Complete product, UX, workflow, data architecture and implementation prompt

Purpose
Build a production-ready, enterprise-grade Trainer Management Web Panel for LoveInStore. The panel must manage the complete employee training lifecycle: discovery, assignment, learning, delivery, attendance, assessment, follow-up, improvement, certification, history and analytics.

Document type: Full build prompt for an AI app builder / product development team
Scope: Responsive web application, role-aware workflows, operational data model and complete interaction states
Version: 1.0

DESIGN PRINCIPLE

Do not build a simple course catalogue or a generic LMS dashboard. Build a Training Operations Management System that connects every training action to a traceable employee journey and measurable field readiness.

0. Instructions to the AI Builder / Development Team

Use this document as the source of truth for the first complete implementation. Build the actual working web panel, not just static mockups. Every visible action must have a meaningful interaction, validation, state transition or clearly labelled integration placeholder.

- **Preserve the business context:** LoveInStore operates in a retail/FMCG field execution environment with employees such as Merchandisers, Promoters, Supervisors, Program Managers and other authorised roles.
- **Do not invent unavailable integrations:** If APIs, authentication, employee master data, notifications, video hosting or file storage are not connected, create a clean adapter/service boundary and realistic demo data. Clearly label demo data.
- **Build end-to-end flows:** Create, edit, view, assign, schedule, conduct, evaluate, follow up, close and inspect historical records.
- **Use consistent entities:** Training Assignment is the central lifecycle record. Employee Training Journey is the consolidated employee-facing history view.
- **Keep records auditable:** Do not silently delete or overwrite completed training, assessment attempts, attendance or follow-up outcomes.
- **Make unresolved business policies configurable:** Program creation permissions, assignment approvals, follow-up ownership and exact employee data scope must be settings/permission policies, not hardcoded assumptions.

Confirmed business decisions

Decision	Confirmed requirement
Employee scope	Trainer can train all relevant employee designations, including Merchandisers, Promoters, Supervisors, Program Managers and other authorised roles.
Training delivery	Support physical, digital/online and hybrid training.
Employee assignment	Trainer can discover eligible employees, select individuals, create batches and assign training.

Important qualification: “All designations” describes role coverage, not unrestricted access to every employee record. Access must still respect the organisation's authorised brand, program, location and data scope.

1. Product Vision and Goals

Create a centralised Training Operations Management System for LoveInStore. It should help Trainers plan, deliver and measure training across multiple employee designations, brands, programs and locations.

Primary goals

- Provide a single operational workspace for Trainers.
- Identify employees who require onboarding, initial training, refresher training or re-training.
- Manage reusable training programs, learning modules and learning resources.
- Assign employees individually or in batches.
- Schedule and conduct physical, digital and hybrid sessions.
- Record attendance, assessment attempts, outcomes and trainer observations.
- Manage follow-ups and practical verification after training.
- Create improvement plans for employees with knowledge or execution gaps.
- Issue and manage certifications when a program requires them.
- Maintain a complete employee-wise and trainer-wise historical record.
- Measure training completion, quality, follow-up effectiveness and employee readiness.

Non-goals

- Do not turn the Trainer Panel into a full HRMS or payroll system.
- Do not allow Trainers to change employee employment status, designation or reporting hierarchy unless separately authorised.
- Do not assume that passing an assessment automatically means an employee is operationally ready.
- Do not treat a batch, a training program and a training session as the same entity.

2. Users, Roles and Permission Model

Primary user: Trainer

- View eligible employees across authorised designations and organisational scope.
- Create and manage training assignments.
- Create batches and schedule sessions.
- Conduct training and record attendance.
- Manage learning resources and assessments, subject to configured permissions.
- Record results, follow-ups, observations and improvement plans.
- View employee training journeys and trainer activity history.
- View training analytics for their authorised scope.

Related roles

Role	Expected responsibility
Trainer	Plans and conducts training; manages assigned employee learning journeys.
Senior Trainer (optional)	May supervise Trainers, review workloads and quality, and access team-level training analytics.
Program Manager	Monitors program execution, employee readiness and operational outcomes; may approve selected actions.
Platform Admin	Controls platform configuration, user access, global settings and authorised data scope.
Employee / Trainee	Receives assignments, attends sessions, accesses permitted learning materials, takes assessments and receives notifications. Employee self-service is an optional connected experience.

Permission matrix

Action	Default Trainer capability
View eligible employees	Yes, within authorised scope
Select employees / create batches	Yes
Assign training	Yes; approval policy configurable
Create/edit training programs	Configurable: Trainer, Program Manager or Admin
Create/edit learning resources	Yes, subject to content permissions
Schedule and manage sessions	Yes
Record attendance	Yes; corrections must be auditable
Create assessments / evaluate	Yes, subject to assessment permissions
Schedule and complete follow-ups	Yes; ownership can be shared/configured
Create improvement plans	Yes
Issue/revoke certificates	Authorisation required
View history	Yes, within authorised scope
Delete completed history	No; use correction or reversal events
Change employee designation/reporting hierarchy	No, unless separately authorised
View financial or unrelated HR data	No, unless separately authorised

Implement permissions as role-based access control (RBAC) combined with scope-based access control. Enforce permissions on the backend/API as well as in the UI. Hiding a button is not sufficient security.

3. Information Architecture and Navigation

Use a left navigation sidebar on desktop, a compact responsive navigation pattern on tablet, and a mobile navigation drawer or equivalent. Keep the active module, page title, breadcrumbs and primary action clear.

Navigation item	Sub-sections / views
Overview	Dashboard, KPI summary, recent activity, upcoming tasks
My Employees	All Employees, New Joiners, Untrained, Training Assigned, In Progress, Completed, Needs Attention, Employee Profiles
Training Programs	All Programs, Create Program, Modules, Assignments, Versions, Archived
Learning Resources	Documents, Videos, Presentations, SOPs, Product Materials, Resource Categories
Batches & Sessions	Batches, Session Calendar, All Sessions, Create Session, Attendance
Assessments	Question Bank, Assessment Builder, Assigned Assessments, Attempts, Results, Reattempts
Follow-ups	Today's Follow-ups, Upcoming, Overdue, Completed, Follow-up History
Improvement & Re-training	Knowledge Gaps, Improvement Plans, Re-training Assignments, Verification
Certifications	Issued, Pending, Expiring, Revoked, Templates
Communication	Announcements, Notifications, Reminders, Message History

Navigation item	Sub-sections / views
Training History	Employee-wise History, Session History, Assessment History, Trainer Activity Log, Audit Events
Reports & Analytics	Completion, Attendance, Assessment, Follow-up, Readiness, Effectiveness, Trainer Productivity
My Profile & Settings	Profile, Availability, Preferences, Notification Settings, Access Scope (read-only if admin-managed)

Global interface elements

- Global search across employees, assignments, programs, batches and sessions, restricted by permissions.
- Notification centre with unread count and actionable notifications.
- Date range and organisational filters where relevant.
- Consistent status badges and status descriptions.
- Quick-create actions for assignment, program, batch, session, assessment and follow-up, where permitted.
- User profile menu, help/support and sign-out.
- Empty, loading, error, permission-denied and no-results states for every major module.

4. Dashboard Requirements

KPI cards

KPI	Definition / calculation
Assigned Employees	Distinct employees with active training assignments in selected scope and date range.
Employees Trained	Distinct employees with at least one completed training assignment in the selected period.
Training Completion	Completed eligible assignments divided by assignments due in the selected period; document exclusions.
Sessions Scheduled	Upcoming sessions in the selected period.
Sessions Completed	Sessions with finalised completion status.
Assessment Pass Rate	Passed evaluated attempts divided by evaluated attempts; show attempt-level and employee-level versions separately.
Follow-ups Due	Open follow-ups due today or earlier.
Overdue Follow-ups	Open follow-ups whose due date/time has passed.
Employees Requiring Attention	Employees with overdue training, failed assessment, unresolved improvement plan or overdue follow-up.
Training Readiness	Employees meeting the configured readiness criteria divided by employees assessed for readiness.

Dashboard components

- Training activity trend over time.
- Training status distribution.
- Upcoming session calendar/list.
- Today's follow-up queue and overdue alerts.
- Employees requiring attention.

- Recent trainer activity timeline.
- Program-wise and designation-wise training summary.
- Quick actions: Assign Training, Create Program, Schedule Session, Create Assessment, Add Follow-up.

Every KPI must open a filtered detail view. Provide date range, brand, program, designation and location filters where permitted. Do not display fabricated analytics as live data.

5. Employee Management and Employee Training Profile

Employee directory

- Search by employee name, employee ID and available contact identifiers.
- Filter by designation, brand, program, region, state, city, territory, supervisor/manager, joining date, employee status and training status.
- Support sorting, pagination, saved filters and bulk selection.
- Show clear eligibility and access-scope reasons when an employee cannot be assigned.
- Support bulk assignment only after a review/confirmation step.
- Prevent accidental duplicate active assignments for the same employee and program/version unless an explicit re-enrolment or repeat-training reason is selected.

Employee profile tabs

Tab	Content
Overview	Employee identity, designation, organisational assignment, manager, location and current training status.
Training Journey	Chronological timeline of assignment, modules, sessions, attendance, assessments, follow-ups, improvement plans and certifications.
Assignments	Active, completed, cancelled and archived training assignments.
Sessions & Attendance	Session-wise attendance, lateness, absence reasons and corrections.
Assessments	Assessment results, attempts, answer review where permitted, feedback and reattempts.
Follow-ups	Upcoming, overdue and completed follow-ups with outcomes and evidence.
Improvement Plans	Knowledge gaps, action items, owners, target dates and verification outcomes.
Certifications	Issued certificates, status, validity and verification details.
Notes & Documents	Trainer remarks, approved attachments and relevant evidence.
Activity Log	Who performed which action and when, with audit details.

Employee profile actions

- Assign training; add to batch; schedule follow-up; assign re-training; view history; export training record.
- Add a note or observation with visibility controls.
- View employee's current training readiness and the exact criteria still outstanding.
- Do not expose unrelated sensitive HR or personal information.

6. Training Program Management

A Training Program defines what an employee needs to learn, why it is required, who it applies to and how completion is measured.

Program fields

- Program ID, name, description and category.
- Brand, client, project or business program association, where applicable.
- Applicable employee designations and target audience.
- Training objectives and expected learning outcomes.
- Prerequisites and eligibility conditions.
- Program owner and authorised trainers.
- Estimated duration and recommended sequence.
- Delivery modes: physical, digital or hybrid.
- Learning modules and linked resources.
- Mandatory/optional module rules.
- Assessment requirements, passing criteria and attempt limits.
- Follow-up and practical verification requirements.
- Certification eligibility and validity rules.
- Effective date, version, status and archive reason.

Program lifecycle

```
Draft -> Review (optional) -> Published -> Active -> Updated / Superseded -> Archived
```

Version programs. When content or pass criteria materially change, create a new version. Existing assignments must retain the program version they were assigned so historical results remain interpretable.

Program management actions

- Create, edit, duplicate, publish, pause, archive and view program history.
- Preview employee experience and module sequence.
- Assign to individual employees or batches.
- View enrolled employees, completion, assessment outcomes and effectiveness.
- Export program-level reports.

7. Learning Modules and Resource Library

Learning module capabilities

- Create modules and arrange them in a sequence.
- Set module objectives, estimated duration and completion criteria.
- Link one or more learning resources.
- Mark modules mandatory or optional.
- Set prerequisites and dependencies.
- Track module started, viewed, completed and verified states where supported.
- Allow module-level quizzes or knowledge checks.

Supported resources

- PDF, PPT/PPTX, DOC/DOCX and approved office documents.
- Training videos and recorded sessions through an approved media provider.
- Product catalogues, brand guidelines, SOPs and reference images.
- External learning links, where permitted.

- Downloadable materials with access controls.

Resource metadata

Store title, description, type, category, tags, owner, file/media reference, version, language, visibility, linked programs/modules, upload date, review date, status and change history. Validate file type and size, scan uploads where supported, and avoid exposing private file URLs.

8. Training Assignment — Central Lifecycle Entity

Training Assignment is the most important transactional entity. It links one employee to one training program version and records the employee's progress through the complete lifecycle.

Assignment fields

- Assignment ID; employee ID; trainer ID/owner; program ID and version.
- Assignment source (manual, bulk, program rule, onboarding or re-training).
- Assignment date, due date, priority and reason.
- Delivery plan and expected completion criteria.
- Batch association, if any.
- Approval status, if approval policy is enabled.
- Current lifecycle status and progress.
- Completion date and final outcome.
- Cancellation/withdrawal reason, if applicable.
- Linked sessions, attendance, assessments, follow-ups, improvement plans and certificates.

Assignment lifecycle

```
Draft -> Pending Approval (optional) -> Assigned -> In Progress -> Assessment Pending -> Follow-up Pending -> Completed / Training Ready
```

Support additional outcomes such as Failed, On Hold, Cancelled, Withdrawn, Expired and Re-training Required. Keep the overall assignment status distinct from individual session, assessment, follow-up and certification statuses.

Assignment rules

- Allow multiple assignments per employee across different programs.
- Prevent duplicate active assignments for the same program/version unless an authorised repeat reason is supplied.
- Preserve prior assignment history when a new assignment is created.
- Support due dates, reminders, priority and overdue escalation.
- Record who assigned, changed, approved or cancelled the assignment.
- If an employee changes brand, program or location, retain historical records and recalculate future eligibility without rewriting past facts.

9. Batch Management

A batch is a group of employees organised for a shared training program or delivery plan. It is not the same as a session.

Batch capabilities

- Create, edit, duplicate, close and archive batches.
- Set batch name, code, program/version, trainer, mode, capacity, start/end dates and location.
- Add/remove eligible employees with validation and audit records.
- Show batch roster, employee status, attendance, completion and assessment summary.

- Support employees joining late or missing an individual session.
- Allow a batch to have multiple sessions and a session to contain a permitted subset of batch members.
- Track batch-level progress without replacing employee-level assignment records.

10. Training Session Management

Session creation fields

- Session ID, title, linked program/module, batch and participant list.
- Session type: classroom, online, field/practical, self-learning review or hybrid.
- Trainer/facilitator, date, start time, end time and timezone.
- Physical venue/address or online meeting provider/link.
- Capacity, required materials and session agenda.
- Pre-session instructions, attendance rules and completion criteria.

Session operations

- Create, reschedule, cancel, start, pause and complete sessions.
- Notify participants and record notification status.
- Mark attendance: Present, Absent, Late, Excused or Not Recorded.
- Capture absence reason and supporting note where required.
- Record session notes, questions, feedback, observations and attachments.
- Record actual start/end time and actual duration.
- Allow authorised attendance correction with reason and audit event.
- Generate a session summary and participant outcome list.

Calendar views

Provide calendar, agenda and list views with filters for trainer, program, batch, delivery mode, status and date range. Prevent or warn about trainer/session scheduling conflicts and capacity violations.

11. Physical, Digital and Hybrid Delivery

Mode	Required support
Physical	Venue, attendance, facilitator, on-site materials, practical demonstration and optional evidence.
Digital	Meeting link or LMS content, online attendance where integration exists, learning progress and digital assessment.
Hybrid	Combination of digital modules and physical sessions, with independent completion tracking for each required component.

Do not claim that online attendance, video watch completion or meeting participation is verified unless a connected provider returns that data. Provide manual verification with an audit trail where automated integration is unavailable.

Mode-specific workflow

```
Program mode -> Session mode -> Participant instructions -> Attendance / content progress -> Completion validation -> Assessment
```

12. Attendance Management

- Record attendance per employee per session.
- Support Present, Absent, Late, Excused and Not Recorded.

- Record check-in/check-out times when applicable.
- Capture absence reasons and trainer remarks.
- Calculate attendance percentage using configurable rules.
- Support make-up sessions for missed training.
- Show employee, batch and program attendance history.
- Prevent attendance records from being silently overwritten.
- Require correction reason and actor identity for edits after session finalisation.

Attendance and training completion are separate. A participant may attend a session but still have incomplete modules, assessment or practical verification.

13. Assessment and Examination Management

Question bank

- Support MCQ single-select, multiple-select, true/false, descriptive and image-based questions.
- Store question text, answer options, correct answer/rubric, explanation, difficulty, topic, tags, language and version.
- Allow reuse across assessments while preserving the question version used in a historical attempt.
- Restrict access to answer keys and evaluation rubrics.

Assessment builder

- Create assessment linked to a program, module, session or assignment.
- Configure duration, availability window, attempt limit, passing score and randomisation.
- Configure negative marking only when explicitly enabled.
- Support automatic scoring for objective questions and manual evaluation for descriptive questions.
- Define retake rules and feedback visibility.
- Preview and publish assessments.

Assessment attempt lifecycle

```
Assigned -> Started -> Submitted -> Auto / Manual Evaluation -> Passed / Failed -> Feedback -> Reattempt (if allowed)
```

Assessment result requirements

- Store each attempt separately; never overwrite earlier attempts.
- Store score, maximum score, percentage, result, start/submission/evaluation timestamps and evaluator.
- Show attempt number and previous-attempt comparison.
- Identify topic-level knowledge gaps.
- Support reattempt scheduling and trainer feedback.
- Record appeals or manual score corrections as auditable events.

14. Follow-up Management

Follow-ups are a first-class operational module and must be linked to a specific Training Assignment. A follow-up is not merely a generic task.

Follow-up record fields

- Follow-up ID, employee, training assignment, related program and owner.
- Follow-up type: understanding check, knowledge retention, field/practical verification, performance review, document check or refresher check.

- Scheduled date/time, due date, method and priority.
- Checklist and expected evidence.
- Employee feedback, trainer observation and outcome.
- Attachment/evidence references.
- Next action and next follow-up date.
- Completion timestamp, completed by and audit history.

Follow-up views

View	Behaviour
Today's Follow-ups	Due today; prioritise by time and urgency.
Upcoming	Future follow-ups, grouped by date.
Overdue	Past due and not completed; show age and escalation state.
Completed	Finalised follow-ups with outcomes and evidence.
All Follow-ups	Searchable/filterable master list.

Follow-up outcomes

- Satisfactory.
- Improvement Required.
- Additional Training Required.
- Unable to Contact.
- Reschedule Required.
- Not Applicable, with reason and authorised closure.

A completed follow-up is immutable as a historical event. If a correction is needed, append a correction event. If the outcome requires more work, create a new follow-up or improvement action rather than reopening history without trace.

15. Improvement Plans and Re-training

Triggers

- Assessment failure or repeated low scores.
- Incomplete mandatory learning.
- Insufficient attendance.
- Unsuccessful practical verification.
- Repeated follow-up concerns.
- A program or SOP update requiring re-certification.

Improvement plan fields

- Plan ID, employee, originating assignment and identified knowledge/skill gap.
- Problem description, evidence and severity/priority.
- Required learning modules and corrective actions.
- Responsible trainer and target dates.
- Re-training sessions and reassessment requirements.
- Verification checklist and final outcome.

- Closure reason, date and approver if required.

Improvement lifecycle

```
Gap Identified -> Plan Created -> Actions Assigned -> Re-training -> Reassessment -> Practical Verification -> Closed / Further Action
```

Do not delete or replace the original failed assessment or follow-up. Link the improvement plan to the original event and preserve the full sequence.

16. Certification Management

- Create certificate templates and configure program-level eligibility.
- Generate unique certificate IDs and downloadable PDF certificates.
- Store employee, program/version, issue date, expiry date, issuer and verification status.
- Support Pending, Issued, Expiring, Expired, Revoked and Renewed statuses.
- Support renewal and re-certification workflows.
- Require an authorised role to revoke or manually issue certificates.
- Provide a verification page or internal verification workflow, subject to privacy rules.
- Retain certificate versions and revocation reasons.

Certification is optional and program-configurable. Passing an assessment should issue a certificate only when all configured requirements are satisfied.

17. Communication and Notifications

Notification events

- New training assignment.
- Assignment due date approaching or overdue.
- Session scheduled, changed or cancelled.
- Session reminder.
- Assessment assigned, due, submitted, evaluated or failed.
- Reattempt available or scheduled.
- Follow-up scheduled, due or overdue.
- Improvement plan created or updated.
- Certification issued or approaching expiry.
- New or updated learning material.

Channels

Support in-app notifications first. Add email, SMS, WhatsApp or push notifications only through approved integrations and configured consent/policy. Keep delivery status, provider reference, timestamp and failure reason.

Communication controls

- Template-based messages with safe variable substitution.
- Preview before bulk announcements.
- Recipient count and audience review before sending.
- Delivery history and retry policy.
- Avoid duplicate reminders and notification spam.

18. Training History, Audit Trail and Employee Journey

Employee-wise Training Journey

Display a chronological timeline combining all events related to the employee's training assignments. Each event should open its source record.

- Training assignment created/updated/approved.
- Program and module completion.
- Session scheduled/rescheduled/cancelled/completed.
- Attendance recorded or corrected.
- Assessment assigned, attempted, evaluated and reattempted.
- Follow-up scheduled and completed.
- Knowledge gap and improvement plan created.
- Re-training conducted.
- Practical verification completed.
- Certificate issued, renewed, expired or revoked.

Trainer-wise activity history

- Programs created/updated/published by the Trainer.
- Employees assigned and batches created.
- Sessions conducted and attendance recorded.
- Assessments created/evaluated.
- Follow-ups completed and outcomes recorded.
- Improvement plans and re-training managed.
- Certificates issued where authorised.

Audit event fields

Event ID, entity type, entity ID, event type, actor ID and role, timestamp, source, previous value, new value, reason, linked records and metadata. Use append-only audit events for critical actions. Protect audit logs from ordinary Trainer modification.

19. Reports and Analytics

Report	Key measures
Employee Training Completion	Assigned, started, completed, overdue and cancelled assignments.
Brand / Program Training	Employee coverage, completion and readiness by brand/program.
Designation / Location	Training status by employee role, region and territory.
Batch Performance	Roster, attendance, completion and assessment outcomes.
Session Report	Scheduled vs completed, attendance, duration and feedback.
Assessment Report	Pass/fail, score distribution, attempt count and knowledge gaps.
Follow-up Report	Due, completed, overdue, outcome and closure time.
Re-training Report	Employees requiring re-training, reason, attempts and improvement.
Certification Report	Issued, pending, expiring, expired and revoked.
Trainer Productivity	Assignments managed, sessions conducted, follow-ups completed and workload.
Training Effectiveness	Pre/post knowledge change and verified practical readiness, where data exists.

Analytics principles

- Every metric must have a documented definition and date basis.
- Distinguish assignment-level, session-level, attempt-level and unique-employee counts.
- Support date range and permitted organisational filters.
- Show denominators and excluded records in rates.
- Do not use pass rate as the only trainer performance measure.
- Allow export to CSV/XLSX/PDF where authorised.
- Provide drill-down from chart/KPI to source records.

20. End-to-End Core Workflows

Workflow A — New employee onboarding

1. Trainer opens Employee Directory and filters new joiners.
2. Trainer reviews employee role, brand/program, location and eligibility.
3. Trainer selects one employee or multiple employees.
4. Trainer selects the applicable onboarding program/version.
5. System checks prerequisites, duplicates and assignment policy.
6. Trainer creates assignment or submits it for approval if required.
7. System notifies employee and updates the Training Journey.
8. Trainer schedules batch/session or assigns digital learning.
9. Attendance/module completion is recorded.
10. Assessment is completed and evaluated.
11. Follow-up and practical verification are scheduled.
12. Employee is marked Training Ready only when configured requirements are met.

Workflow B — Existing employee refresher training

1. Identify employees due for refresher training through filters, expiry rules or program updates.
2. Review previous training version, scores, attendance and follow-up outcomes.
3. Create a new assignment linked to the current program version.
4. Assign required modules and session(s).
5. Conduct training and assessment.
6. Complete follow-up and update readiness.

Workflow C — Failed assessment

1. Assessment attempt is evaluated as Failed.
2. System records score, attempt number and topic-level gaps.
3. Trainer reviews result and provides feedback.
4. If attempts remain, Trainer schedules a reattempt or corrective learning.
5. If further intervention is needed, create an Improvement Plan.
6. Conduct re-training, reassessment and verification.
7. Preserve every attempt and outcome in the Training Journey.

Workflow D — Practical follow-up

1. Trainer opens Follow-ups and selects a due activity.
2. Trainer reviews linked assignment and expected checklist.
3. Trainer contacts or visits the employee using the selected method.
4. Trainer records employee feedback, observations and evidence.
5. Trainer selects outcome.
6. If satisfactory, close the follow-up and update readiness criteria.
7. If improvement is required, create a linked action, next follow-up or Improvement Plan.

Workflow E — Hybrid training

1. Program contains digital modules and physical practical session requirements.
2. Trainer assigns the program and creates the delivery plan.
3. Employee completes digital content and required knowledge checks.
4. Trainer conducts physical demonstration/practical session.
5. Attendance and practical verification are recorded.
6. Final assessment and follow-up are completed.
7. System evaluates all required components before marking the assignment complete/ready.

21. Status Model and State Transitions

Object	Suggested statuses
Training Program	Draft, In Review (optional), Published, Active, Paused, Superseded, Archived
Training Assignment	Draft, Pending Approval, Assigned, In Progress, Assessment Pending, Follow-up Pending, Completed, Training Ready, Failed, On Hold, Cancelled, Withdrawn, Re-training Required
Batch	Draft, Planned, Active, Completed, Closed, Cancelled, Archived
Session	Draft, Scheduled, Rescheduled, In Progress, Completed, Cancelled, No-show/Not Conducted
Attendance	Not Recorded, Present, Late, Absent, Excused
Assessment Attempt	Assigned, Started, Submitted, Under Evaluation, Passed, Failed, Invalidated (authorised only)
Follow-up	Scheduled, Due, In Progress, Completed, Overdue, Rescheduled, Cancelled
Improvement Plan	Open, In Progress, Awaiting Verification, Closed, Escalated, Cancelled
Certification	Pending, Issued, Expiring, Expired, Revoked, Renewed

Use explicit state-transition rules. Prevent impossible transitions, require reasons for cancellations/reversals and record every state change.

22. Data Model and Relationships

Implement a normalised relational data model. The following entities are the minimum recommended foundation; add supporting tables for attachments, notification delivery, templates, settings and integrations as required.

Entity	Purpose / important relationships
Trainer	Trainer identity, profile, role, active status and authorised scope.
Employee	Employee master reference, designation and organisational attributes.
TrainingProgram	Program definition and lifecycle.
TrainingProgramVersion	Immutable versioned curriculum and rules.
LearningModule	Module sequence, objectives and completion rules.
LearningResource	File/media/link metadata, version and permissions.
ProgramModuleResource	Many-to-many link between program/module and resource.
TrainingAssignment	Central employee-to-program-version lifecycle record.
TrainingBatch	Group of participants for a common delivery plan.
BatchMember	Membership linking batch and employee/assignment.
TrainingSession	Scheduled or completed event linked to program/batch.
SessionParticipant	Participant list for each session.
AttendanceRecord	Employee attendance for a session.
Assessment	Assessment definition and configuration.
Question	Reusable, versioned question bank item.
AssessmentQuestion	Question selection/order and assessment-specific settings.

Entity	Purpose / important relationships
AssessmentAttempt	Immutable attempt and result record.
AssessmentAnswer	Submitted answer and evaluation data.
FollowUp	Scheduled follow-up linked to TrainingAssignment.
ImprovementPlan	Corrective plan linked to an identified gap.
ImprovementAction	Tasks/modules/retraining/verification within a plan.
Certification	Credential and validity record.
TrainingEvent	Chronological domain event for employee journey.
AuditLog	Append-only actor/action/change record.
Notification	Message event, recipient and delivery status.
Attachment	Secure file references and metadata.
ApprovalRequest	Optional approval workflow for assignments/programs/certificates.

Core relationship rules

- One employee can have many Training Assignments.
- One program has many immutable versions; each assignment references exactly one version.
- One assignment can have many sessions, assessment attempts, follow-ups, improvement plans and certificates.
- A batch can contain many assignments; a session can include a subset of batch members.
- Attendance is recorded per employee per session, not only at batch level.
- Assessment attempts are separate records and never overwritten.
- Follow-ups reference the assignment and, where applicable, the exact session or assessment event.
- TrainingEvent supports timeline presentation; AuditLog supports accountability. They are related but not interchangeable.

23. Technical Architecture and Implementation Guidance

Application architecture

- Build a responsive web application with modular frontend components and a clear service/API layer.
- Use a relational database for transactional training records and enforce foreign keys, uniqueness and referential integrity.
- Separate authentication, authorisation, employee directory, training operations, file storage, notification and reporting concerns.
- Use server-side validation for every mutation.
- Use background jobs for reminders, certificate generation, bulk notifications and report exports.
- Use pagination, indexed search and server-side filters for large employee and history tables.
- Use timezone-aware timestamps; display dates in the user's configured timezone while preserving canonical timestamps.
- Use soft archive for configurable master records and append-only events for critical history.
- Add observability: structured logs, error tracking and audit-safe operational metrics.

Suggested service boundaries

Service / domain	Responsibilities
Identity & Access	Login, sessions, RBAC, scope policies and permission checks.
Employee Directory	Employee lookup, filters, organisational attributes and eligibility.
Program & Content	Programs, versions, modules, resources and content access.
Assignment	Employee enrolment, assignment lifecycle and approvals.
Batch & Session	Batch membership, calendars, session operations and attendance.
Assessment	Question bank, assessments, attempts, scoring and results.
Follow-up & Improvement	Follow-up scheduling, outcomes, improvement plans and verification.
Certification	Eligibility, generation, validity, renewal and revocation.
History & Audit	Employee journey, trainer activity and immutable audit events.
Notification	Templates, delivery providers, reminders and delivery history.
Analytics	Metric definitions, aggregates, drill-down and exports.

Integration points

- LoveInStore employee master / HR source, if available.
- Existing brand, client, program and location hierarchy.
- LoveInStore authentication and role management.
- Approved file storage and media hosting.
- Email/SMS/WhatsApp/push providers, if enabled.
- Online meeting provider or LMS integrations, if required.
- Existing reporting or BI systems, if applicable.

Do not assume any integration exists. Inspect the existing repository and APIs first. Reuse existing authentication, design system, data models and conventions where available. Do not replace or overwrite existing backend systems without explicit instruction.

24. UI/UX and Visual Design Direction

Design principles

- Create a premium, clear, enterprise-grade operations interface.
- Prioritise information hierarchy, readability, density and fast task completion.
- Use restrained colour, consistent spacing, strong typography and accessible contrast.
- Avoid generic AI-dashboard styling, excessive gradients, excessive glassmorphism, neon glow and decorative-only motion.
- Use meaningful charts and status indicators only when they improve decisions.
- Use tables for operational records, timelines for history, calendars for scheduling and focused forms for creation/editing.
- Use consistent page layouts without making every module look like identical repeated cards.

Suggested visual system

Element	Direction
Primary tone	Deep navy / blue for navigation, structure and primary actions.
Background	Neutral light grey/off-white surfaces with clear white content areas.
Typography	Modern sans-serif such as Inter; strong hierarchy and readable data labels.
Status colours	Green = completed/success; amber = due/attention; red = failed/overdue; blue = in progress/information; grey = inactive/neutral.
Cards	Use for KPIs and high-level summaries, not as a replacement for operational tables.
Tables	Search, sort, filters, pagination, row actions and clear empty/loading states.
Forms	Grouped sections, inline validation, save draft and clear unsaved-change handling.
Motion	Subtle transitions for navigation, state feedback and progressive disclosure; respect reduced-motion preferences.

Responsive behaviour

- Desktop: full sidebar, dense but readable tables, multi-column layouts and calendar views.
- Tablet: collapsible navigation, simplified filter panels and adaptive table columns.
- Mobile: single-column layouts, filter drawer, stacked detail sections and accessible touch targets.
- Do not merely shrink the desktop layout; recompose workflows for narrow screens.
- Keep high-priority actions reachable without hiding important status or context.

Accessibility

- Meet WCAG 2.2 AA as the target.
- Keyboard navigation and visible focus states.
- Semantic headings, labels and form controls.
- Sufficient contrast; do not rely on colour alone to convey status.
- Accessible tables, dialogs, tooltips and validation messages.
- Respect reduced-motion preferences.

25. Forms, Validation and Bulk Operations

- Validate required fields, date/time ordering, capacity, eligibility and duplicate assignments.
- Use confirmation dialogs for bulk assignment, cancellation, certificate revocation and other consequential actions.
- Show selected employee count and a preview of recipients before bulk actions.
- Allow partial success in bulk operations only with a clear row-level result report.
- Provide CSV import only if approved; validate rows before committing and return a downloadable error report.
- Use drafts for long program, assessment and improvement-plan forms.
- Warn about unsaved changes and conflicting session schedules.
- Use idempotency protection for retryable bulk/API operations.

26. Security, Privacy and Data Governance

- Enforce least-privilege access and tenant/organisation scope on every API request.
- Do not expose employee contact details or unrelated HR data unless necessary and authorised.
- Secure uploaded resources and certificates; use expiring or permission-checked access links.
- Validate and scan uploaded files where infrastructure supports it.
- Protect question answer keys and assessment content from unauthorised access.
- Maintain audit logs for assignment, attendance, assessment, follow-up, certificate and permission changes.
- Apply retention and deletion policies according to company policy and applicable law.
- Use secure authentication, session expiry, CSRF protection where applicable, rate limiting and input validation.
- Do not place secrets, tokens or private file URLs in frontend code or logs.
- Support export controls and record who exported sensitive reports.

27. Seed Data and Demonstration Scenario

Provide realistic demo data so the panel can be reviewed end-to-end. Clearly label it as demo/sample data.

- At least 3 Trainers, including one with a senior/supervisory permission set if that role is implemented.
- At least 30 employees across Merchandiser, Promoter, Supervisor and Program Manager designations.
- Multiple brands/programs and locations, including an FMCG example such as Haldiram's.
- At least 5 training programs across onboarding, product, application, operational, compliance and refresher categories.
- Program versions, modules, PDFs/videos placeholders and reusable resources.
- Multiple batches with different sizes and delivery modes.
- Past, current and upcoming sessions with mixed attendance outcomes.
- Assessments with passed, failed, pending, manually evaluated and reattempted examples.
- Follow-ups in upcoming, due, overdue and completed states.
- Improvement plans with open and closed examples.
- Certificates issued, pending, expiring and revoked.
- Employee journey timelines and trainer activity logs.

Required demo story

Create a sample Merchandiser named Rahul Sharma assigned to an FMCG product training program. Show assignment, physical product session, attendance, assessment pass, scheduled follow-up, practical verification and Training Ready status. Also show another employee who fails an assessment, receives an improvement plan, completes re-training and passes a later attempt. Ensure both journeys preserve the full history.

28. Acceptance Criteria

1. Trainer can log in and see only permitted records and actions.
2. Trainer can search/filter eligible employees across authorised designations.
3. Trainer can select employees individually or in bulk and create assignments.
4. System validates duplicate active assignments and eligibility.
5. Trainer can create/manage a program and version according to permissions.
6. Trainer can create learning modules and link reusable resources.
7. Trainer can create batches and schedule multiple sessions.
8. System supports physical, digital and hybrid delivery plans.
9. Trainer can record attendance per employee per session.
10. Trainer can create assessments and evaluate attempts.

11 Every assessment attempt is retained independently.

12 Trainer can create and manage follow-ups linked to assignments.

13 Overdue follow-ups are visible and actionable.

14 Trainer can create improvement plans and assign re-training.

15 Training Ready is only reached when configured criteria are satisfied.

16 Certificates are issued only when eligibility rules are satisfied and the actor is authorised.

17 Employee profile shows a complete chronological Training Journey.

18 Trainer profile/history shows attributable activity.

19 Reports support filtering and drill-down to source records.

20 Important state changes and corrections are auditable.

21 All primary actions work with validation, feedback and appropriate error states.

22 Responsive layouts work on desktop, tablet and mobile.

23 Accessibility and security requirements are implemented and tested.

29. Required Screens / Page Inventory

Area	Screens to build
Authentication	Login, forgot password, session expired, access denied
Dashboard	Trainer overview, KPI drill-down, recent activity
Employees	Directory, filters, bulk selection, employee profile, training journey
Programs	Program list, create/edit wizard, program detail, version history, module builder
Resources	Resource library, upload/edit, preview, category management
Batches	Batch list, create/edit, batch detail, roster
Sessions	Calendar, session list, create/edit, session detail, attendance, session summary
Assessments	Question bank, question editor, assessment builder, assigned assessments, attempt detail, result review
Follow-ups	Today's, upcoming, overdue, completed, follow-up detail/create/edit
Improvement	Knowledge gap queue, improvement plan list, plan detail, re-training assignment, verification
Certifications	Certificate list, detail, template management, expiry and revocation workflow
Communication	Notification centre, announcement composer, delivery history
History	Employee journey, trainer activity, session history, audit event detail
Reports	Report catalogue, filters, charts, drill-down, export
Settings	Profile, preferences, notification settings, permission/scope view

Build all critical screens and their connected interactions. Do not stop after producing the dashboard or a handful of static pages.

30. Final Build Instructions

1 Inspect the existing LoveInStore project/repository, framework, design system, authentication and API conventions before implementation.

2. Prepare a concise implementation plan and identify existing components/services to reuse.
3. Implement the database schema and migrations for the entities in this specification, adapting to existing platform conventions.
4. Implement backend/API validation, permission checks, lifecycle transitions and audit events.
5. Build the responsive UI and complete all critical user journeys.
6. Add realistic demo/seed data and make the application usable without external integrations.
7. Connect integrations only when credentials, endpoints and approved provider details are available.
8. Test all acceptance criteria, permission boundaries, edge cases and responsive layouts.
9. Provide a final summary of implemented modules, routes, data entities, integrations, known limitations and test results.

Do not silently omit features. If a feature cannot be fully implemented because an API, provider or business rule is missing, implement the UI and domain boundary, mark it as awaiting configuration, and list the exact dependency. Do not fabricate a successful integration.

Appendix A — Complete High-Level Workflow

```
TRAINER LOGIN
|
v
DASHBOARD
|
v
EMPLOYEE DIRECTORY --> FILTER / SELECT EMPLOYEES
|
v
CREATE INDIVIDUAL ASSIGNMENT OR BATCH
|
v
SELECT PROGRAM + VERSION + LEARNING MODULES
|
v
CHOOSE DELIVERY: PHYSICAL / DIGITAL / HYBRID
|
v
SCHEDULE SESSIONS + NOTIFY PARTICIPANTS
|
v
CONDUCT TRAINING / COMPLETE DIGITAL LEARNING
|
v
RECORD ATTENDANCE + MODULE COMPLETION
|
v
ASSESSMENT --> PASS? ---------------- NO
| |
YES v
| KNOWLEDGE GAP
v |
TRAINING COMPLETED v
| IMPROVEMENT PLAN
v |
FOLLOW-UP / PRACTICAL VERIFICATION v
| RE-TRAINING
v |
SATISFACTORY? ----------------------------+
|
YES
v
CERTIFICATION (IF REQUIRED)
|
v
TRAINING READY
|
v
UPDATE EMPLOYEE JOURNEY + TRAINER HISTORY + ANALYTICS
```

Appendix B — Open Business Policies to Configure

The following decisions were not yet confirmed. Implement them as configurable policies and do not hardcode a single assumption:

Policy	Recommended configurable options
Who creates training programs?	Trainer; Program Manager/Admin only; or both with approval/version controls.
Does assignment require approval?	No approval; Program Manager approval; conditional approval by brand/program.
Who owns follow-ups?	Trainer; Program Manager; shared with Supervisor/Manager; configurable by program.
What employee records can a Trainer access?	All within authorised organisation; assigned programs; assigned locations; or combined brand/program/location scope.

Recommended initial default for a demo: Trainer can create and manage programs within authorised scope; assignments do not require approval; Trainer owns training follow-ups while Supervisors/Managers can contribute verification; employee access is constrained by combined authorised brand/program/location scope. Treat these as demo defaults only until the business confirms them.

Appendix C — Final Product Quality Checklist

- Is Training Assignment the central lifecycle record?
- Are program versions immutable for historical assignments?
- Can one employee have multiple independent training journeys?
- Can one batch have multiple sessions and partial attendance?
- Are attendance and completion distinct?
- Are assessment attempts preserved individually?
- Are follow-ups linked to assignments and outcomes?
- Can a failed employee complete a traceable improvement journey?
- Is Training Ready based on explicit criteria?
- Can the Trainer answer who trained whom, on what, when, how, with what result and what follow-up?
- Are all actions permission-checked and auditable?
- Do the dashboard and reports drill down to real records?
- Do desktop, tablet and mobile flows remain usable?
- Are unavailable integrations clearly identified rather than simulated as live?